CHAPTER 3

[Part 4]

PROBABILITY THEORY

Conditional Probability

- Let A and B be events, and assume that P(B)>0.

- The conditional probability of A given B is,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Definition: The **conditional probability** of an event **A** in relationship to an event **B** is the probability that event **A** occurs given that event **B** has already occurred. The notation for conditional probability is **P(A|B)** [pronounced as ***the probability of event A given B***].

Example 18

Question: Suppose that we roll 2 fair dice. What is the probability of getting a sum of 10?

Solution:

Sample space

(1,1)	(2,1)	(3,1)	(4,1)	(5,1)	(6,1
(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	(6,2
(1,3)	(2,3)	(3,3)	(4,3)	(5,3)	(6,3
(1,4)	(2,4)	(3,4)	(4,4)	(5,4)	(6,4
(1,5)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5
(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6

- The probability of getting a sum

Example 19

Question: Refer to **Example 18**, what is the probability of getting a sum of 10 given that at least one die is 5?

Solution:

- Let A denote the event "getting a sum of 10".
- Let B denote the event "at least one die is 5".
- Thus, $P(A \cap B) = 1/36$ and $P(B) = 11/36$
- The probability of getting a sum of 10 given that at least 1 die is 5 is,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/36}{11/36} = \frac{1}{11}$$

Example 20

Suppose that we roll 2 fair dice. Find the probability of **getting a sum of 7, given that the digit in the first die is greater than in the second.**

Solution

- The sample space S consists of 6× outcomes.

- Let A be the event "the sum of di the 2 dice is 7"

$A=\{(6,1),(5,2),(4,3),(3,4),(2,5),($

Solution (cont.)

- Let B be the event "the digit in th
  die is greater than the second"

$B=\{ (6,1), (6,2), (6,3), (6,4), (6,5), (\cdot$
$(5,2), (5,3), (5,4), (4,1), (4,2), (\cdot$
$(3,1), (3,2), (2,1) \}$

$P(B)=15/36$

Solution (cont.)

- Let C be the event "the sum of digit the 2 dice is 7 but the digit in the firs is greater than the second"

$$C=\{ (6,1), (5,2), (4,3) \} = A \cap B$$

$$P(A \cap B)= 3/36$$

Solution (cont.)

- The probability of getting a sum given that the digit in the first d greater than in the second is,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{(3/36)}{(15/36)}$$

Example 21

Weather records show that the probability of high barometric pressure is 0.85 and the probability of rain and high barometric pressure is 0.15. What is the probability of rain given high barometric pressure?

Solution:

- Let ***H*** denote the event "rain".
- Let ***T*** denote the event "high barometric pressure".
- The probability of rain given high barometric pressure is,

$$P(H|T) = \frac{P(H \cap T)}{P(T)} = \frac{0.15}{0.85} = 0.1765$$

Exercise #4

The probability that a doctor correctly diagnose a particular illness is 0.7. Given that the doctor makes an incorrect diagnosis, the probability that the patient files a lawsuit is 0.9. What is the probability that the doctor makes an incorrect diagnosis and the patient sues?

Solution

Let,

A: doctor correctly diagnose

B: patient files a lawsuit

Given, $P(A) = 0.7$; $P(B|A') = 0.9$

Thus, $P(A' \cap B) = P(B|A')P(A') = 0.9 \times 0.3 = 0.27$

Example 24

Paediatric department researcher examines the medical records of toddlers that came to a particular paediatric clinic. He found that 20% of them came for flu treatment and 10% of mothers of the toddler that having flu are also having flu. 30% of the mothers that came to the clinic are found having flu.

a) What is the probability of the toddler having flu given that the mother having flu.

b) What if the probability of the toddler having flu given that the mother is not having flu.

Solution

a) What is the probability of the toddler having flu given that the mother having flu.

Let, T : event of toddler having flu; M: event of mother having flu

$$P(T)=0.2;\ \ P(M\cap T)=0.1; P(M)=0.3$$

$$P(T\mid M)=\frac{P(M\cap T)}{P(M)}=\frac{0.1}{0.3}=0.33$$

a) What if the probability of the toddler having flu given that the mother is not having flu.

According to the law of total probability:

$$P(T)=P(T\cap M)+P\left(T\cap M'\right);\ \text{where}\ \ P(M)+P(M')=1$$

$$0.2=0.1+P\left(T\cap M'\right)$$

$$\therefore P\left(T\cap M'\right)=0.2-0.1=0.1;\ \ P(M')=1-P(M)=1-0.3=0.7$$

$$\therefore P(T\mid M')=\frac{P(T\cap M')}{P(M')}=\frac{0.1}{0.7}=0.143$$